

Image quality

When you consider buying a printer, the image quality it offers is sure to be uppermost on your mind. If the test of a printer is in its ability to print image documents that are faithful to the original, then its ability to print documents with text and images combined may be considered the acid test. We divided the image quality test into two parts:

The combi-document: This document, as mentioned earlier, contained both high-resolution images as well as text, which were combined in a manner that would comprehensively test the printer's image reproduction capabilities, including its ability to replicate the finer details of the original.

The Flower-Woman photo: This was the second part of the test procedure wherein the image (see *image in the Scanner comparison test*) contained several minute details, such as the wrinkles on the woman's forehead, details in the clothing, contrast and sharpness in the flowers, etc that would put the printer to the test.

Image quality (combi-document)

We used photo quality inkjet paper, and the document was printed in photo or high-resolution mode as opposed to the normal or draft mode that was used in the speed test. We observed the four-colour checkered box in the output to evaluate the colour depth and the sharpness of the checks. Only five printers passed this test without any errors—the Epson Stylus Color 870, the Epson Stylus Color 880, the Canon BJC 5100, the Canon BJC 7100, and the Wipro Emerge e7460. Seven of the printers displayed an almost flawless performance—for instance, the HP DJ 930C lost a point for pixelation in the black checks, while the Epson Stylus Color 680 lost a point for uneven checks.

The concentric circles image, which had lines drawn from the centre to the circumference of the outermost circle, was checked for accuracy of reproduction. The Epson Stylus Photo 870 secured the highest score allocated for reproducing this feature, while the others lost out because of jagged lines or non-uniform colouring.

Next, we observed the seven coloured

bars with 10 levels of colour saturation to determine each printer's ability to capture the colour variations. The winner in this department was the **Hewlett-Packard DJ 990Cxi**—it was able to accurately reproduce 59 of the 70 coloured



HP DeskJet 990Cxi: High in quality but also in price

boxes. The same vendor's DJ 930C came a close second by flawlessly picking up 56 of the 70 boxes, while the Epson Stylus Photo 870 took third place by picking up 55 boxes.

The next section we observed was the black box with the yellow text. Here again, the Hewlett-Packard printers took the top positions and were followed closely by the Epson Stylus Color 880 and the Kodak Personal Picture Maker 200. The Lexmark range fared the poorest, with all the outputs displaying a splurge of black without even a tint of the yellow colour of the text in the original.

We also considered a passport-sized photograph of the flower-woman to determine the printer's ability to pick out

minute details even from such a small image. The joint winners here were the **Hewlett-Packard DJ 990Cxi** and the **Kodak PPM 200**, with the Epson Stylus Color 880 and three other HP printers losing out narrowly.

Finally, we determined the printer's ability to pick up small font sizes by observing the reproduction of the text printed in four different font sizes. This is a relevant feature if you want to print multiple pages in a single sheet: You would then have to depreciate the font size and would want your printouts to still be readable.

All the Lexmark printers barring the Z12, as well as HP's 990Cxi, 970Cxi, 930C, and 950C, were able to print three of the font sizes. Others that matched this performance were the Kodak PPM 200, the Epson Stylus Photo 870, and the Canon BJC 7100.

We then aggregated the scores achieved by each printer. The overall winners were the Canon BJC 7100, and Hewlett-Packard's DeskJet 950C, 840C and 930C. The other impressive performers were the Lexmark Z52, the Hewlett-Packard DJ 990Cxi, and the Kodak PPM 200.

Image quality (Flower-woman)

We concluded the image quality test by observing the printouts of the Flower Woman photo, and this was the most crucial part of the test as the image contained numerous minute details.

We checked for faithful reproduction

Inkjet Maintenance

Here are a few tips on maintaining your inkjet printer:

Aligning/calibrating the print head: This is one of the most important tasks to be performed after installing a new printer or a new cartridge. If the print head is not calibrated properly, it may result in abnormal or bad print quality. The newest range of printers from Hewlett-Packard have an automatic calibration process wherein the print head is calibrated automatically after the printer is installed or when the printer detects a cartridge change.

Cleaning the print head: If you encounter problems while printing, such as abnormal linings, bandings, uneven ink flow, or missing colours, the printer nozzles are probably clogged with excess dried ink that needs to be cleaned. This

can be done from the toolbox or the monitoring software that gets installed along with the printer drivers. Generally, a test pattern is printed while the cleaning process is on, and this can help you determine whether a particular cartridge is out of ink.

Checking for paper jams: This is a frequent problem with inkjet printers, and could happen because of dust collecting in the paper path. You have to clean the printer regularly, especially the area near the paper output tray. Check for bits of paper and even stapler or paper pins that could cause paper jams and eventually damage the printer's assembly. It is a good practice to use a blower or vacuum cleaner once in a while to remove dust from the parts that are not accessible with your bare hands.

